

Task 1 Report

Supervised Probing of Embeddings

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TABLE OF CONTENTS

INTRODUCTION	1
1 DATA AND PREPROCESSING	3
1.1 MISSING DATA ASSESSMENT	3
1.2 COHORT DEFINITION AND EMBEDDING EXTRACTION	4
1.3 TARGET CLASS DISTRIBUTION	4
1.4 PCA VISUALIZATION OF EMBEDDING SPACES	6
2 EVALUATION PROTOCOL	11
2.1 CROSS-VALIDATION STRATEGY	11
2.2 EVALUATION METRICS	11
2.3 FEATURE SCALING	11
2.4 EVALUATION PIPELINE	12
2.5 REPEATED EXPERIMENTS	12
3 SIMPLE MODELS	13
3.1 SUMMARY OF RESULTS	14
3.2 INTERPRETATION	15
4 HYPERPARAMETER TUNING	16
4.1 BEST CONFIGURATIONS	17
4.2 INTERPRETATION	17
5 ENSEMBLE MODELS	19
5.1 SUMMARY OF RESULTS	20
5.2 INTERPRETATION	21
5.3 HYPERPARAMETER TUNING AND FINAL SELECTION	21

6	NEURAL NETWORK (MLP)	23
6.1	LEARNING CURVES	24
6.2	SUMMARY OF RESULTS	25
6.3	INTERPRETATION	25
6.4	SYSTEMATIC HYPERPARAMETER OPTIMIZATION FOR THE MLP	26
7	STATISTICAL VALIDATION OF REPRESENTATION LEVELS	27
8	FINAL SUMMARY TABLE	28
8.1	CLASS-SPECIFIC ERROR ANALYSIS (CONFUSION MATRIX)	28
8.2	MAIN FINDINGS	31
8.3	COMPARISON OF REPRESENTATION LEVELS	32
8.4	ANALYSIS OF STAGE PREDICTION	33
	CONCLUSION	35

INTRODUCTION

The objective of Task 1 is to evaluate how well the patient embeddings (Level 0-3) capture clinically useful information. To do this, I used embeddings as inputs for supervised classifiers and measured their ability to predict three clinical targets: PAM50 molecular subtype (5 classes), survival risk (binary), and stage (5 classes).

The performance of the embeddings is assessed using three metrics: macro F1-score, balanced accuracy, and AUC. All experiments were repeated 10 times using different random seeds, and results are reported as mean $\pm$ standard deviation to ensure robustness.

From a clinical perspective, the targets selected for this study are particularly relevant. The PAM50 molecular subtypes are widely used in breast cancer management to guide therapeutic decisions and estimate prognosis. Survival status provides direct prognostic information, while pathological stage characterises disease extent and tumour progression. Using these variables as references therefore provides an indirect measure of the clinical relevance of the learned representations.

From a methodological perspective, the incorporation of protein–protein interaction (PPI) networks as an inductive bias is a distinctive aspect of this project. The underlying assumption is that integrating prior biological knowledge during representation learning, either through graph regularisation (Level 2) or through graph-based message passing within a GNN encoder (Level 3) encourages the emergence of biologically meaningful and clinically informative structures within the embedding space.

I formulate the following hypotheses on the progression of clinical signal across levels:

- Level 1 (denoising autoencoder) > Level 0 (simple autoencoder): denoising should improve the robustness of representations by reducing the influence of experimental noise.
- Level 2 (PPI graph regularisation) > Level 1(DAE): integrating biological knowledge from the protein-protein interaction network should structure the latent space in a more biologically coherent way. By bringing genes involved in similar processes closer together in the representation, molecular subtypes may become more easily separable.
- Level 3 (GNN encoder using the PPI graph): explicit use of the graph may capture more complex biological dependencies and improve the prediction of global phenotypes such as survival. However, propagating information between neighbouring genes may also lead to over-smoothing, making the representations of connected genes more similar. In that case,

fine-grained molecular signatures, which may be important for tasks such as PAM50 classification, could be partially attenuated.

This report describes the data preprocessing, experimental protocol, results for simple models, hyperparameter tuning, ensemble models, and the MLP. A final comparison and conclusion are also provided.

1 DATA AND PREPROCESSING

For each representation level, the embedding matrix was merged with the clinical dataset using *patient_id* from the embedding files and *patient_id_short* from the PAM50 dataset (a new column I created to match the identifiers). Following this merge, the resulting dataset contained 953 patients and 89 variables.

1.1 MISSING DATA ASSESSMENT

Missing values were assessed after merging the datasets. The pattern of missingness across clinical variables is summarized below.

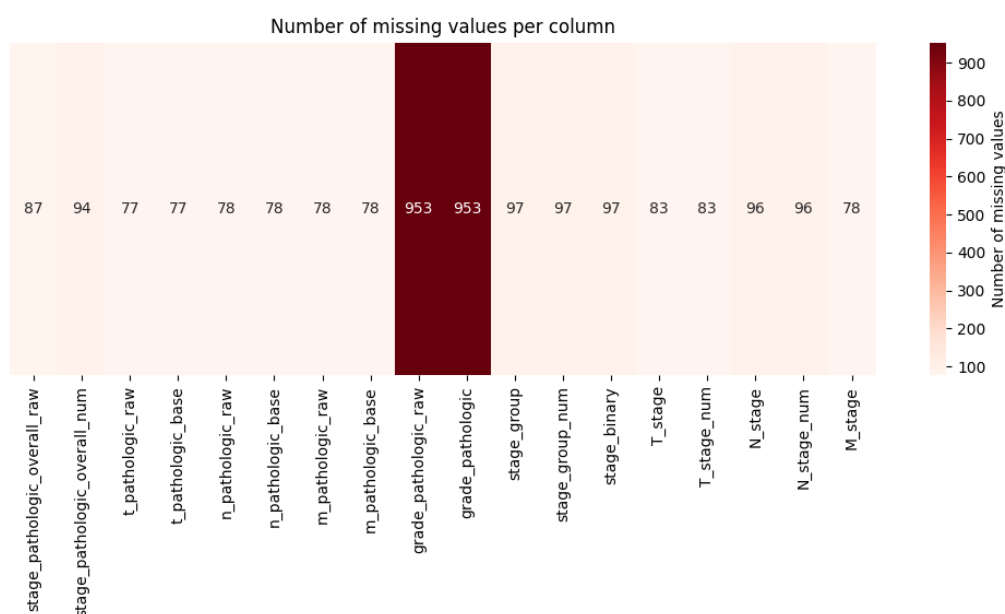


Figure 1: Number of missing values

Most missing values were observed in TNM-related variables (*T_stage_num*, *N_stage_num*, *M_stage*), with between 77 and 96 missing observations depending on the variable. These variables are not used in Task 1 but will be retained for subsequent analyses in Task 2. For statistical analyses such as chi-square tests, samples with missing values will be excluded when necessary.

Two variables related to tumour grade (*grade_pathologic_raw* and *grade_pathologic*) contained missing values for all 953 patients and were therefore excluded from further analysis.

1.2 COHORT DEFINITION AND EMBEDDING EXTRACTION

After preprocessing, PAM50 subtype labels and survival information were available for all 953 patients. In contrast, the pathological stage variable (*stage_pathologic_overall_num*) contained 94 missing values, resulting in a cohort of 859 patients for stage prediction.

For each representation level, patient embeddings were extracted to serve as input features for the downstream classification models. The embedding dimensionality remained strictly constant across all levels, with a latent space of 64 features per patient.

The final cohort sizes and corresponding embedding shapes for each independent clinical target are summarized below.

Target	Number of Patients	Embedding Shape
PAM50	953	(953, 64)
Survival	953	(953, 64)
Stage	859	(859, 64)

The stage prediction task was based on the variable *stage_pathologic_overall_num*, which contains five categories (Stage 0-IV).

1.3 TARGET CLASS DISTRIBUTION

To facilitate the interpretation of classification performance, class distributions were examined for each prediction task.

PAM50 Molecular Subtypes

Class	Number of patients	Percentage
Luminal A	294	30.8%
Luminal B	214	22.5%
Basal-like	214	22.5%
HER2-enriched	130	13.6%
Normal-like	101	10.6%

The PAM50 dataset is moderately imbalanced, with Luminal A representing the largest subgroup.

Stage

Class	Number of patients	Percentage
Stage 1	153	16.1%
Stage 2	509	53.4%
Stage 3	177	18.6%
Stage 4	17	1.8%
Stage 0	3	0.3%

The stage prediction task is highly imbalanced, with Stage II accounting for more than half of all patients, whereas Stage 0 and Stage IV are strongly underrepresented. This strong imbalance partly explains the low F1-scores observed for stage prediction.

Survival

Class	Number of patients	Percentage
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Low Risk (0)	476	49.9%
High Risk (1)	477	50.1%

The survival dataset is almost perfectly balanced between the two classes.

1.4 PCA VISUALIZATION OF EMBEDDING SPACES

To obtain an initial qualitative view of the learned representations, a Principal Component Analysis (PCA) was applied to the embeddings of each level. The 64-dimensional embeddings were projected onto the first two principal components and coloured according to PAM50 subtype, survival, and stage.

The resulting projections are shown in Figure 2 and Figure 3.

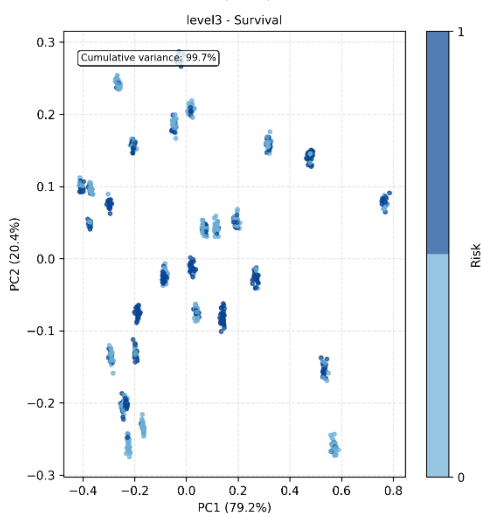
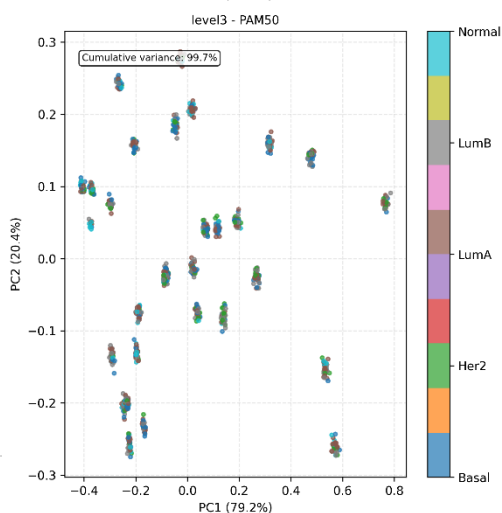
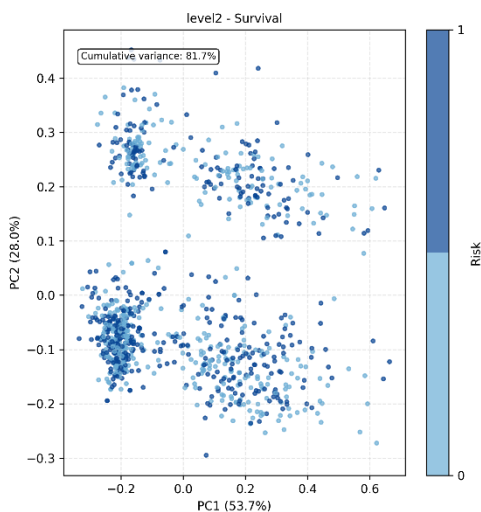
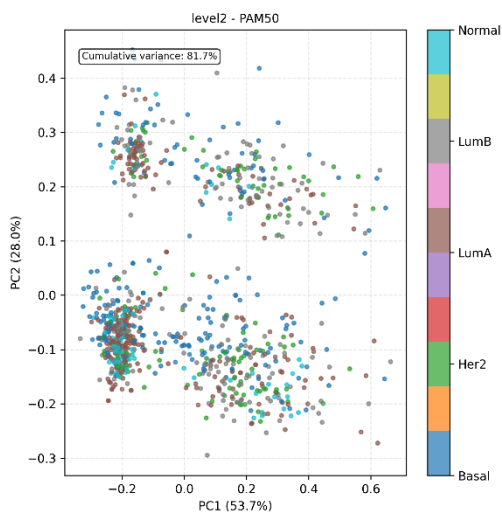
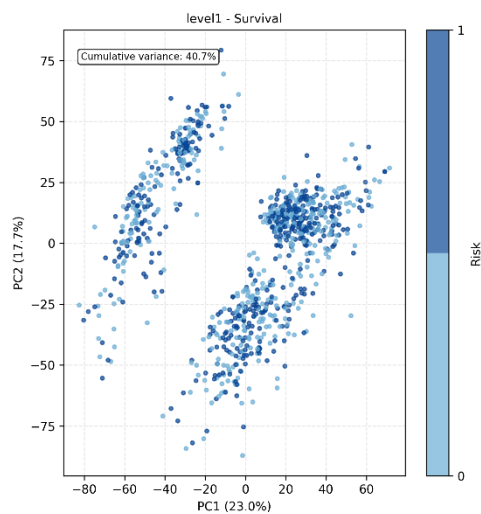
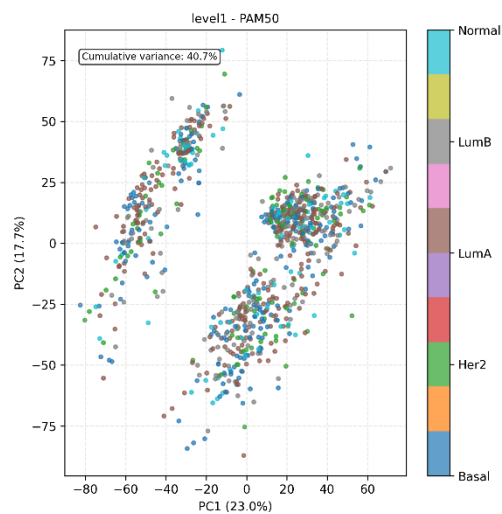
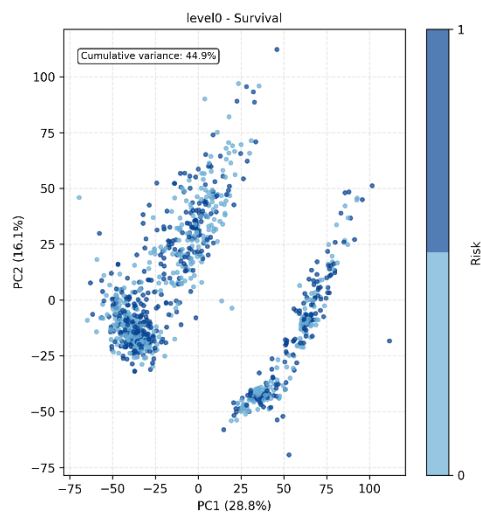
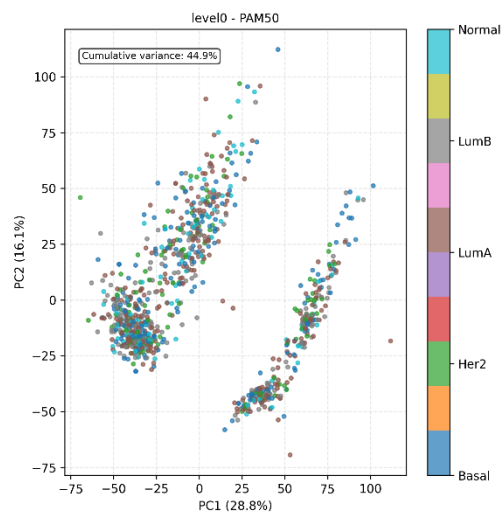


Figure 2 : PCA projections for PAM50 and Survival across Levels 0 to 3.

PCA projections for PAM50 (left column) and survival (right column) across Levels 0 to 3. Each row corresponds to a representation level. The percentage of variance explained by each component and the cumulative variance are shown in each subplot.

Level 0 (AE) already exhibits a structured latent space with several visible patient groups. However, PAM50 subtypes and survival labels remain strongly intermixed, suggesting that clinically relevant information is not directly separable in the first two principal components.

The denoising procedure (Level 1, DAE) modifies the geometry of the latent space and produces more compact patient distributions. Nevertheless, no clear visual separation of the clinical labels is observed. This suggests that any potential improvement in predictive performance is likely to be modest.

Compared with Levels 0 and 1, Level 2 (GR-DAE) exhibits a more structured organisation of patients. Several regions of the latent space appear enriched for specific PAM50 subtypes, although substantial overlap remains. This visual organisation suggests that Level 2 embeddings may facilitate the separation of PAM50 molecular subtypes, a hypothesis that will be assessed quantitatively in the supervised probing experiments. In contrast, survival labels remain largely mixed.

Level 3 (GS-DAE) displays a markedly different geometry characterised by numerous compact clusters. This behaviour is consistent with the graph propagation mechanism of the GNN encoder. PAM50 classes remain highly intermixed, suggesting that fine-grained subtype information may be less accessible in the first two principal components. Interestingly, survival groups are also not clearly separated in the two-dimensional projection. This indicates that, if prognostic information is present, it may be encoded in higher-dimensional structures that are not fully captured by the PCA representation.

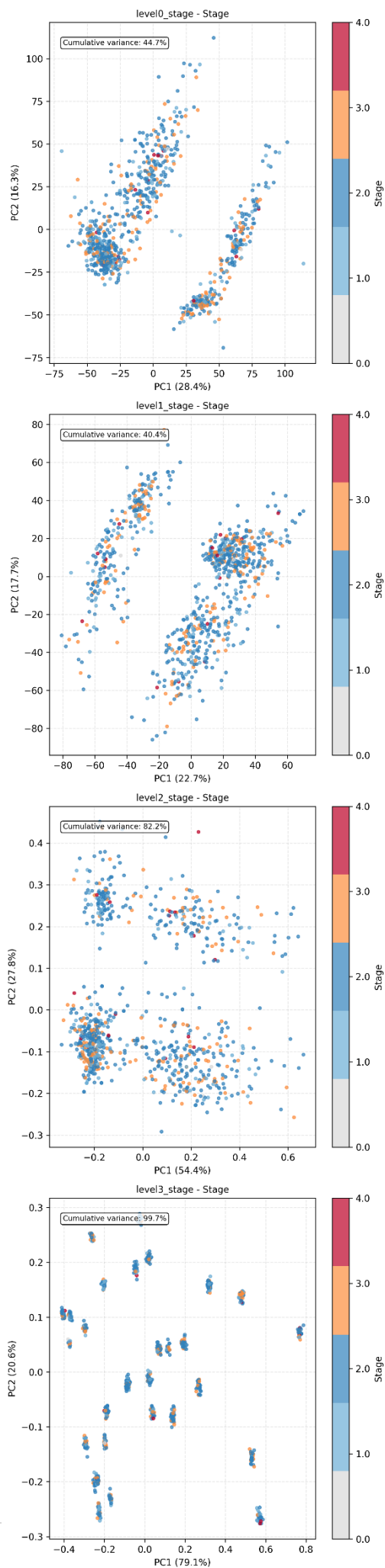


Figure 3 : PCA projections for stage across Levels 0 to 3.

For stage, no clear separation of stage groups is observed for any representation level, reflecting the difficulty of capturing stage information across all tasks.

Overall, the PCA projections illustrate how the geometry of the latent space evolves across representation levels. While they provide useful qualitative insights, they only capture a small fraction of the information contained in the original 64-dimensional embeddings.

2 EVALUATION PROTOCOL

The quality of the patient embeddings was assessed through supervised prediction of the three clinical targets: PAM50 subtype, survival, and stage.

2.1 CROSS-VALIDATION STRATEGY

A stratified cross-validation framework was adopted using *StratifiedKFold* to ensure a fair comparison between representation levels. For PAM50 and survival prediction, a 5-fold stratified cross-validation was implemented using *StratifiedKFold* with shuffling enabled and a fixed random seed (*random_state* = 42). The same train-test splits were used across all embedding levels and classifiers to guarantee comparability of the results.

For stage prediction, some classes contained very few samples, particularly Stage 0 ($n = 3$) and, to a lesser extent, Stage IV ($n = 17$). As a result, a 5-fold stratified cross-validation could not guarantee adequate class representation in every fold. Therefore, a 3-fold stratified cross-validation was used.

2.2 EVALUATION METRICS

Three complementary metrics were used to evaluate classifier performance:

- **Macro F1-score:** suitable for imbalanced classes, as it averages the F1-score of each class without favouring the majority class.
- **Balanced accuracy:** averages the recall per class, which is more reliable than standard accuracy when classes are imbalanced.
- **AUC (one-vs-rest):** adapted to multi-class classification by comparing each class against all others, then averaging the results. For binary classification, the probability of the positive class is used instead.

2.3 FEATURE SCALING

Several classifiers considered in this study, including Logistic Regression, Support Vector Machines (SVM), k-Nearest Neighbours (k-NN), and Multilayer Perceptrons (MLP), are sensitive to feature

scaling. Consequently, the embedding vectors were standardized using *StandardScaler* within each training fold before model fitting.

Tree-based methods, including Random Forest, XGBoost, and AdaBoost, do not rely on distance calculations and therefore were evaluated without feature standardisation.

2.4 EVALUATION PIPELINE

A common evaluation pipeline was implemented for all representation levels, target variables, and classifiers. For each cross-validation fold, the pipeline:

1. Fits a *StandardScaler* on the training data when required;
2. Trains the classifier on the training set;
3. Generates predictions on the validation set;
4. Computes Macro F1-score, Balanced Accuracy, and AUC.

Performance estimates were obtained by averaging the metrics across all folds.

2.5 REPEATED EXPERIMENTS

To reduce the impact of random variation during model training, each experiment was repeated ten times using different random seeds. Final results are reported as mean $\pm$ standard deviation across repetitions.

3 SIMPLE MODELS

Three baseline classifiers were evaluated:

- **Logistic Regression** (*random_state=42, max_iter=200*)
- **Linear SVM** (*kernel='linear', probability=True, random_state=42*)
- **k-Nearest Neighbors** (*k=5*)

These models were selected because they are simple, interpretable, and commonly used to probe the quality of learned representations. Each classifier was evaluated on all embedding levels (Levels 0–3) and all prediction tasks (PAM50, survival, and stage) using the cross-validation protocol described previously.

The complete results (mean $\pm$ standard deviation across 10 repetitions) are reported in the table below. Corresponding visualisations are provided in the barplot.

Target	Level	Classifier	F1_mean	F1_std	BalAcc_mean	BalAcc_std	AUC_mean	AUC_std
PAM50	level0	Logistic Regression	0.1869	0.0049	0.2001	0.0035	0.4929	0.0055
PAM50	level0	SVM (linear)	0.1716	0.0081	0.2007	0.0083	0.4903	0.0077
PAM50	level0	k-NN (k=5)	0.1916	0.0106	0.1983	0.0102	0.4957	0.0067
PAM50	level1	Logistic Regression	0.1987	0.001	0.2103	0.0086	0.5138	0.0077
PAM50	level1	SVM (linear)	0.1813	0.0081	0.2055	0.0068	0.4989	0.0081
PAM50	level1	k-NN (k=5)	0.2317	0.0124	0.2354	0.0118	0.5282	0.0066
PAM50	level2	Logistic Regression	0.3566	0.0061	0.3789	0.0051	0.7580	0.0026
PAM50	level2	SVM (linear)	0.3492	0.0094	0.3773	0.0071	0.7588	0.0035
PAM50	level2	k-NN (k=5)	0.3480	0.0161	0.3551	0.0145	0.6718	0.0092
PAM50	level3	Logistic Regression	0.1439	0.0060	0.1984	0.0044	0.5309	0.0065
PAM50	level3	SVM (linear)	0.1047	0.0039	0.1983	0.0026	0.5378	0.0069
PAM50	level3	k-NN (k=5)	0.2221	0.0064	0.2287	0.0062	0.5429	0.0049
Survival	level0	Logistic Regression	0.5024	0.0097	0.5035	0.0092	0.5005	0.0084
Survival	level0	SVM (linear)	0.4975	0.0091	0.4993	0.0083	0.4934	0.0126
Survival	level0	k-NN (k=5)	0.5166	0.0119	0.5176	0.0120	0.5173	0.0108
Survival	level1	Logistic Regression	0.5026	0.0150	0.5037	0.0155	0.5011	0.0128
Survival	level1	SVM (linear)	0.4969	0.0143	0.4992	0.0150	0.4927	0.0117
Survival	level1	k-NN (k=5)	0.5040	0.0124	0.5050	0.0122	0.5082	0.0076

Survival	level2	Logistic Regression	0.5050	0.0098	0.5064	0.0096	0.52	0.0086
Survival	level2	SVM (linear)	0.4924	0.0105	0.5000	0.0099	0.5010	0.0110
Survival	level2	k-NN (k=5)	0.5066	0.0073	0.5074	0.0072	0.5182	0.0077
Survival	level3	Logistic Regression	0.6404	0.0073	0.6411	0.0072	0.7062	0.0045
Survival	level3	SVM (linear)	0.6182	0.0105	0.6197	0.0096	0.701	0.0102
Survival	level3	k-NN (k=5)	0.6905	0.0077	0.6910	0.0077	0.7462	0.0055
Stage	level0_stage	Logistic Regression	0.1806	0.0090	0.2047	0.0060	0.4326	0.0069
Stage	level0_stage	SVM (linear)	0.1645	0.0049	0.2038	0.0040	0.5535	0.0234
Stage	level0_stage	k-NN (k=5)	0.2032	0.0111	0.2120	0.0085	0.5133	0.0083
Stage	level1_stage	Logistic Regression	0.1785	0.0057	0.2011	0.0034	0.5606	0.0139
Stage	level1_stage	SVM (linear)	0.1491	0.0013	0.1980	0.0010	0.4691	0.0185
Stage	level1_stage	k-NN (k=5)	0.2164	0.0112	0.2251	0.0092	0.5153	0.0102
Stage	level2_stage	Logistic Regression	0.1491	0.0010	0.1976	0.0010	0.5867	0.0082
Stage	level2_stage	SVM (linear)	0.1488	0.0	0.2	0.0	0.5258	0.0250
Stage	level2_stage	k-NN (k=5)	0.1917	0.0065	0.1994	0.0063	0.5188	0.0074
Stage	level3_stage	Logistic Regression	0.1490	0.0006	0.1997	0.0007	0.6188	0.0104
Stage	level3_stage	SVM (linear)	0.1488	0.0	0.2	0.0	0.5099	0.0323
Stage	level3_stage	k-NN (k=5)	0.1990	0.0101	0.2083	0.0086	0.5345	0.0123

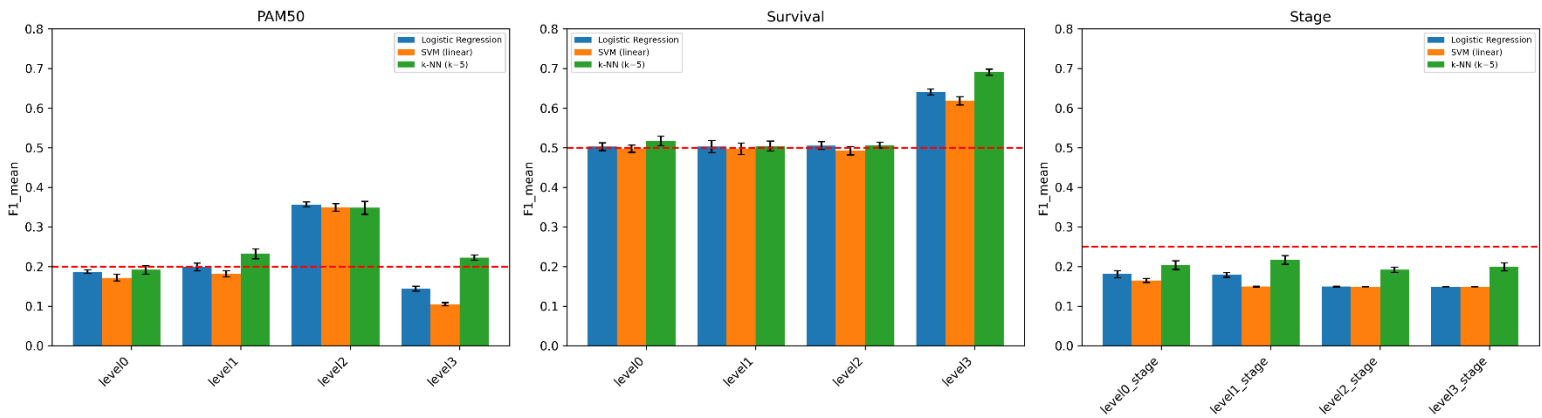


Figure 4: Barplot of simple models

3.1 SUMMARY OF RESULTS

For PAM50 classification, the best performance was obtained at Level 2(GR-DAE). Logistic Regression achieved the highest Macro F1-score (0.357 ± 0.006), closely followed by Linear SVM (0.349 ± 0.009) and k-NN (0.348 ± 0.016).

For survival prediction, Level 3(GS-DAE) substantially outperformed all other representation levels. The best result was achieved by k-NN, reaching a Macro F1-score of 0.691 ± 0.008 and an AUC of 0.746 ± 0.006 .

Stage prediction proved considerably more challenging. Across all embedding levels and classifiers, Macro F1-scores remained below 0.22.

Overall, standard deviations were low, indicating stable performance across repeated runs.

3.2 INTERPRETATION

The results reveal distinct patterns across the three clinical targets.

For PAM50 classification, performance improves markedly from Level 0 to Level 2, with the best F1-score increasing from approximately 0.19 to 0.36. However, performance drops substantially at Level 3(GS-DAE). This observation is consistent with the hypothesis that graph propagation may attenuate fine-grained molecular signatures that are important for distinguishing PAM50 subtypes. One possible explanation is an over-smoothing effect, whereby neighbouring genes acquire increasingly similar representations and subtype-specific information becomes less discriminative.

A different pattern is observed for survival prediction. Levels 0–2 perform close to random classification ($F1 \approx 0.50$ and $AUC \approx 0.50$), whereas Level 3(GS-DAE) produces a substantial improvement across all metrics. This suggests that the GNN-based representation captures prognostic information that is not readily accessible in the lower-level embeddings. Such information may arise from the integration of biological context through the PPI network and the aggregation of signals distributed across multiple genes.

Finally, stage prediction remains difficult regardless of the representation level. The consistently low F1-scores may partly be explained by the strong class imbalance observed in the dataset, particularly the very small number of Stage 0 and Stage IV samples. These results suggest that the current embeddings contain limited information for accurate stage discrimination, although the impact of class imbalance should also be taken into account.

Based on these results, Level 2(GR-DAE) was selected for hyperparameter tuning for PAM50, and Level 3(GS-DAE) for survival.

4 HYPERPARAMETER TUNING

Following the baseline experiments, a manual hyperparameter exploration was conducted on the best-performing embedding levels identified previously: Level 2(GR-DAE) for PAM50 classification and Level 3(GS-DAE) for survival prediction.

The following parameters were investigated:

- **k-NN:** ($k \in \{3,5,7,9,11,15\}$)
- **SVM:** $\text{kernel} \in \{\text{linear}, \text{polynomial}, \text{RBF}\}$, ($C \in \{0.1,1,10\}$)
- **Logistic Regression:** ($C \in \{0.1,1,10\}$)

The complete results are reported in the table below.

Target	Model	Param	F1_mean	F1_std	BalAcc_mean	BalAcc_std	AUC_mean	AUC_std
PAM50	k-NN	k=3	0.3316	0.0154	0.3428	0.0144	0.6473	0.0083
PAM50	k-NN	k=5	0.3479	0.0160	0.3551	0.0144	0.6717	0.0092
PAM50	k-NN	k=7	0.3407	0.0113	0.3506	0.0107	0.6862	0.0050
PAM50	k-NN	k=9	0.3449	0.0170	0.3538	0.0152	0.6962	0.0063
PAM50	k-NN	k=11	0.3468	0.0139	0.3571	0.0129	0.7058	0.0052
PAM50	k-NN	k=15	0.3379	0.0148	0.3496	0.0131	0.7116	0.0051
PAM50	SVM	kernel=linear C=0.1	0.3006	0.0071	0.3336	0.0049	0.7206	0.0031
PAM50	SVM	kernel=linear C=1	0.3491	0.0094	0.3772	0.0070	0.7588	0.0035
PAM50	SVM	kernel=linear C=10	0.3933	0.0092	0.4153	0.0084	0.7933	0.0031
PAM50	SVM	kernel=poly C=0.1	0.1988	0.0048	0.2532	0.0034	0.6959	0.0026
PAM50	SVM	kernel=poly C=1	0.2803	0.0079	0.3141	0.0062	0.7187	0.0025
PAM50	SVM	kernel=poly C=10	0.3489	0.0118	0.3780	0.0097	0.7422	0.0048
PAM50	SVM	kernel=rbf C=0.1	0.2208	0.0079	0.2681	0.0078	0.7061	0.0034
PAM50	SVM	kernel=rbf C=1	0.3253	0.0056	0.3589	0.0064	0.7411	0.0023
PAM50	SVM	kernel=rbf C=10	0.3709	0.0101	0.3927	0.0095	0.7724	0.0027
PAM50	LR	C=0.1	0.3189	0.0064	0.3428	0.0053	0.7203	0.0025
PAM50	LR	C=1	0.3565	0.0060	0.3789	0.0050	0.7579	0.0026
PAM50	LR	C=10	0.4332	0.0095	0.4367	0.0078	0.7921	0.0030
Survival	k-NN	k=3	0.6936	0.0073	0.6941	0.0072	0.7315	0.0053
Survival	k-NN	k=5	0.6905	0.0077	0.6910	0.0077	0.7462	0.0055
Survival	k-NN	k=7	0.6877	0.0082	0.6883	0.0081	0.7543	0.0066

Survival	k-NN	k=9	0.6858	0.0059	0.6864	0.0060	0.7602	0.0057
Survival	k-NN	k=11	0.6879	0.0043	0.6885	0.0042	0.7643	0.0045
Survival	k-NN	k=15	0.6950	0.0098	0.6955	0.0095	0.7701	0.0050
Survival	SVM	kernel=linear C=0.1	0.5760	0.0052	0.5767	0.0053	0.6129	0.0056
Survival	SVM	kernel=linear C=1	0.6182	0.0105	0.6197	0.0096	0.7017	0.0102
Survival	SVM	kernel=linear C=10	0.7051	0.0035	0.7055	0.0034	0.7633	0.0031
Survival	SVM	kernel=poly C=0.1	0.57	0.0043	0.5732	0.0036	0.6021	0.0075
Survival	SVM	kernel=poly C=1	0.5931	0.0087	0.6029	0.0082	0.6647	0.0059
Survival	SVM	kernel=poly C=10	0.60	0.0047	0.6207	0.0049	0.7096	0.0044
Survival	SVM	kernel=rbf C=0.1	0.6108	0.0064	0.6183	0.0057	0.6951	0.0036
Survival	SVM	kernel=rbf C=1	0.6695	0.0053	0.6701	0.0052	0.7382	0.0036
Survival	SVM	kernel=rbf C=10	0.6848	0.0063	0.6863	0.0062	0.7610	0.0067
Survival	LR	C=0.1	0.5805	0.0073	0.5811	0.0071	0.6302	0.0040
Survival	LR	C=1	0.6404	0.0073	0.6411	0.0072	0.7062	0.0045
Survival	LR	C=10	0.6917	0.0039	0.6924	0.0037	0.7540	0.0043

4.1 BEST CONFIGURATIONS

For PAM50 classification, the best performance was achieved by Logistic Regression with ($C = 10$), reaching a Macro F1-score of 0.433 ± 0.010 , substantially improving upon the baseline result (0.357 ± 0.006).

For survival prediction, the highest Macro F1-score was obtained with a Linear SVM ($(C = 10)$), achieving 0.705 ± 0.004 . Logistic Regression with ($C = 10$) achieved a very similar performance (0.692 ± 0.004), while k-NN reached 0.695 ± 0.010 for ($k = 15$).

4.2 INTERPRETATION

Increasing the regularisation parameter (C) consistently improved the performance of both Logistic Regression and Linear SVM for PAM50 and survival prediction. This suggests that the default level of regularisation was overly restrictive and that more discriminative decision boundaries could be learned from the embeddings.

For SVMs, the linear kernel consistently outperformed polynomial and RBF kernels across both prediction tasks. Interestingly, Logistic Regression achieved the best overall PAM50 performance, while Linear SVM achieved the best survival performance.

These observations suggest that a large fraction of the clinically relevant information encoded in the embeddings is already linearly accessible. In other words, complex non-linear decision boundaries appear to provide limited additional benefit compared with well-regularised linear models.

Overall, the results support the initial objective of Task 1, namely evaluating how much clinically useful information is encoded within the learned representations and can be recovered using relatively simple predictive models.

5 ENSEMBLE MODELS

In addition to the baseline classifiers, three ensemble learning methods were evaluated:

- Random forest ($n_estimators = 100$)
- XGBoost ($eval_metric = 'logloss'$)
- AdaBoost ($n_estimators = 100$)

Default hyperparameters were used for all ensemble models. The complete results are reported in the table below, with corresponding visualisations provided in the barplot.

Target	Level	Classifier	F1_mean	F1_std	BalAcc_mean	BalAcc_std	AUC_mean	AUC_std
PAM50	level0	Random Forest	0.1970	0.01	0.2154	0.0101	0.5077	0.0089
PAM50	level0	XGBoost	0.2020	0.0095	0.2114	0.0093	0.5060	0.0088
PAM50	level0	AdaBoost	0.1712	0.0082	0.2053	0.0067	0.5011	0.0067
PAM50	level1	Random Forest	0.2227	0.0167	0.2296	0.0147	0.5293	0.0099
PAM50	level1	XGBoost	0.2258	0.0073	0.2295	0.0068	0.5322	0.0077
PAM50	level1	AdaBoost	0.1835	0.0096	0.2096	0.0089	0.5086	0.0115
PAM50	level2	Random Forest	0.3783	0.0118	0.3822	0.0111	0.7328	0.0047
PAM50	level2	XGBoost	0.3893	0.0113	0.3910	0.01087	0.7360	0.0055
PAM50	level2	AdaBoost	0.3434	0.0212	0.3490	0.0166	0.7158	0.0049
PAM50	level3	Random Forest	0.2190	0.0082	0.2196	0.0074	0.5397	0.0056
PAM50	level3	XGBoost	0.2267	0.0123	0.2272	0.0117	0.5384	0.0070
PAM50	level3	AdaBoost	0.145	0.0090	0.2017	0.0066	0.5473	0.0063
Survival	level0	Random Forest	0.5025	0.0121	0.5038	0.0118	0.5201	0.0121
Survival	level0	XGBoost	0.5037	0.0083	0.5044	0.0081	0.5087	0.0105
Survival	level0	AdaBoost	0.5039	0.0142	0.5056	0.0138	0.5086	0.0099
Survival	level1	Random Forest	0.5217	0.0106	0.5226	0.0105	0.5254	0.0124
Survival	level1	XGBoost	0.5064	0.0096	0.5069	0.0095	0.5150	0.0118
Survival	level1	AdaBoost	0.5066	0.0096	0.5077	0.0095	0.5166	0.0137
Survival	level2	Random Forest	0.5076	0.0115	0.5082	0.0117	0.5230	0.0117
Survival	level2	XGBoost	0.5041	0.0141	0.5048	0.01408	0.5186	0.0169
Survival	level2	AdaBoost	0.5172	0.0112	0.5203	0.0116	0.5308	0.0157
Survival	level3	Random Forest	0.6850	0.0078	0.6856	0.0078	0.7521	0.0068
Survival	level3	XGBoost	0.6879	0.0072	0.6884	0.0073	0.7523	0.0088
Survival	level3	AdaBoost	0.6867	0.0109	0.6877	0.0108	0.7563	0.0077

Stage	level0_stage	Random Forest	0.2001	0.0082	0.2174	0.0048	0.5032	0.0124
Stage	level0_stage	XGBoost	0.2088	0.0091	0.2171	0.0064	0.5118	0.0225
Stage	level0_stage	AdaBoost	0.1908	0.0134	0.2035	0.017	0.4845	0.0240
Stage	level1_stage	Random Forest	0.1903	0.0068	0.2122	0.0045	0.5101	0.0255
Stage	level1_stage	XGBoost	0.2066	0.0097	0.2151	0.0064	0.4849	0.0205
Stage	level1_stage	AdaBoost	0.1964	0.0076	0.2063	0.0049	0.4390	0.0239
Stage	level2_stage	Random Forest	0.1895	0.0070	0.2042	0.0050	0.5162	0.0224
Stage	level2_stage	XGBoost	0.1981	0.0094	0.2062	0.0084	0.5279	0.0164
Stage	level2_stage	AdaBoost	0.1901	0.0063	0.2030	0.0053	0.5494	0.0162
Stage	level3_stage	Random Forest	0.2159	0.0139	0.2193	0.0119	0.5434	0.0167
Stage	level3_stage	XGBoost	0.2272	0.0157	0.2276	0.0120	0.5746	0.0326
Stage	level3_stage	AdaBoost	0.1741	0.0071	0.2003	0.0038	0.5925	0.0163

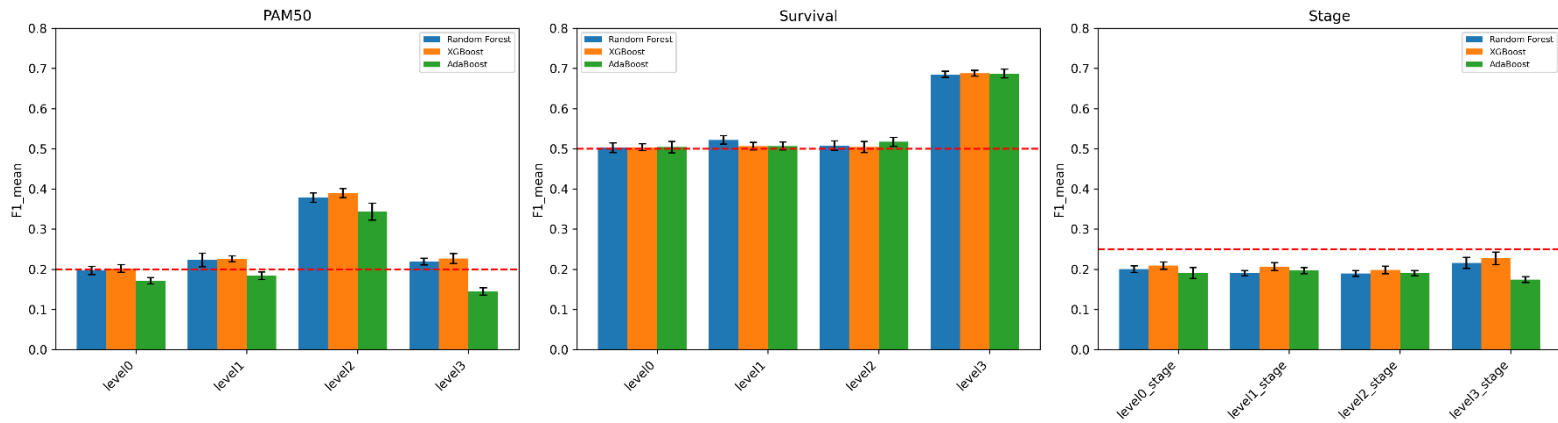


Figure 5: Barplot of ensemble models

5.1 SUMMARY OF RESULTS

For PAM50 classification, the best ensemble performance was achieved by XGBoost on Level 2(GR-DAE), reaching a Macro F1-score of 0.389 ± 0.011 . Although this represents a substantial

improvement over the baseline models, it remains below the performance obtained by the tuned Logistic Regression model (0.433 ± 0.010).

For survival prediction, the best ensemble results were obtained on Level 3(GS-DAE). XGBoost achieved a Macro F1-score of 0.688 ± 0.007 , while Random Forest reached 0.685 ± 0.008 . These performances are comparable to those obtained with the best tuned simple models.

As observed previously, stage prediction remained challenging across all representation levels. The highest Macro F1-score was obtained by XGBoost on Level 3(GS-DAE) (0.227 ± 0.016), still close to the random baseline of 0.25. All scores remained below 0.23.

5.2 INTERPRETATION

Despite their greater modelling flexibility, ensemble methods did not consistently outperform the best simple models. For PAM50 classification, the tuned Logistic Regression model achieved a substantially higher Macro F1-score than all ensemble approaches, including XGBoost. Similarly, for survival prediction, ensemble methods only matched, rather than exceeded, the performance of the best tuned linear and distance-based models.

One possible explanation is that a large proportion of the clinically relevant information encoded in the embeddings is already linearly accessible. This interpretation is supported by the strong performance of Logistic Regression and Linear SVM observed in the previous section. If the latent representations are already well structured, highly flexible models may provide limited additional benefit.

Several other factors may also contribute to these observations. First, the sample size remains relatively limited compared with the complexity of ensemble methods such as Random Forest and XGBoost. Second, stage prediction is affected by severe class imbalance, particularly for Stage 0 and Stage IV, which may hinder the learning process regardless of the classifier used. Finally, ensemble models were evaluated using default hyperparameters; further optimization could potentially improve their performance.

5.3 HYPERPARAMETER TUNING AND FINAL SELECTION

A systematic grid search was conducted for the ensemble models using GridSearchCV with 5-fold cross-validation and a fixed random seed (42).

- **PAM50:** XGBoost (with parameters: *learning_rate* = 0.1, *max_depth* = 6, *n_estimators* = 200) achieved the best ensemble performance (Macro F1 = 0.4075), but remained below the tuned Logistic Regression (Macro F1 = 0.433).
- **Survival:** XGBoost (with parameters: *learning_rate* = 0.3, *max_depth* = 3, *n_estimators* = 100) reached a Macro F1 of 0.7111, slightly outperforming the tuned Linear SVM (Macro F1 = 0.705). Given the marginal difference and the simplicity of linear models, the Linear SVM was retained as the final model for survival prediction.

Overall, these results reinforce the conclusion that the learned embeddings contain clinically relevant information that can be effectively recovered using relatively simple predictive models. More complex ensemble approaches provide only modest gains, suggesting that representation quality may be more important than classifier complexity in this setting.

6 NEURAL NETWORK (MLP)

To evaluate whether more expressive neural models could extract additional information from the embeddings, two multilayer perceptron (MLP) architectures were investigated:

- **Simple MLP:** one hidden layer containing 50 neurons.
- **Tuned MLP:** two hidden layers (128 $\rightarrow$ 64 neurons), L2 regularisation ($\alpha = 0.001$), early stopping, and an adaptive learning rate.

The complete results are reported in the table below.

Target	Level	Classifier	F1_mean	F1_std	BalAcc_mean	BalAcc_std	AUC_mean	AUC_std
PAM50	level0	MLP (simple)	0.1837	0.0098	0.2042	0.0101	0.5022	0.0109
PAM50	level0	MLP (tuned)	0.1852	0.01333	0.2139	0.0091	0.5120	0.0077
PAM50	level1	MLP (simple)	0.1875	0.0119	0.2075	0.0109	0.5098	0.0092
PAM50	level1	MLP (tuned)	0.1916	0.0130	0.2184	0.0084	0.5209	0.0081
PAM50	level2	MLP (simple)	0.2768	0.0157	0.3028	0.0153	0.6752	0.0142
PAM50	level2	MLP (tuned)	0.3336	0.0213	0.3559	0.0187	0.7260	0.0141
PAM50	level3	MLP (simple)	0.1588	0.0095	0.2004	0.0051	0.5270	0.0075
PAM50	level3	MLP (tuned)	0.1559	0.0154	0.2033	0.0061	0.5476	0.0109
Survival	level0	MLP (simple)	0.4897	0.0139	0.5006	0.0182	0.5015	0.0140
Survival	level0	MLP (tuned)	0.5078	0.0113	0.5129	0.0121	0.5146	0.0125
Survival	level1	MLP (simple)	0.4918	0.0179	0.498	0.0186	0.4966	0.0225
Survival	level1	MLP (tuned)	0.5046	0.0144	0.5105	0.0131	0.5112	0.0130
Survival	level2	MLP (simple)	0.4836	0.0161	0.5016	0.0129	0.5064	0.0131
Survival	level2	MLP (tuned)	0.4841	0.0154	0.5013	0.0123	0.5176	0.0118
Survival	level3	MLP (simple)	0.5969	0.0152	0.6012	0.0147	0.6412	0.0143
Survival	level3	MLP (tuned)	0.6710	0.01226	0.6740	0.0116	0.7265	0.0092
Stage	level0_stage	MLP (simple)	0.1845	0.0083	0.2096	0.0039	0.5159	0.0333

Stage	level0_stage	MLP (tuned)	0.1664	0.0077	0.2049	0.0036	0.5242	0.0250
Stage	level1_stage	MLP (simple)	0.1654	0.0073	0.1988	0.0030	0.5476	0.0256
Stage	level1_stage	MLP (tuned)	0.1661	0.0110	0.2035	0.0028	0.5580	0.0188
Stage	level2_stage	MLP (simple)	0.1539	0.0048	0.1980	0.0029	0.5086	0.0220
Stage	level2_stage	MLP (tuned)	0.1487	0.0007	0.1989	0.0012	0.4966	0.0254
Stage	level3_stage	MLP (simple)	0.1577	0.0072	0.2009	0.0017	0.4481	0.0213
Stage	level3_stage	MLP (tuned)	0.1508	0.0032	0.1999	0.0015	0.4524	0.0193

6.1 LEARNING CURVES

Learning curves were analysed for the most relevant configurations: PAM50 prediction using Level 2(GR-DAE) embeddings and survival prediction using Level 3(GS-DAE) embeddings.

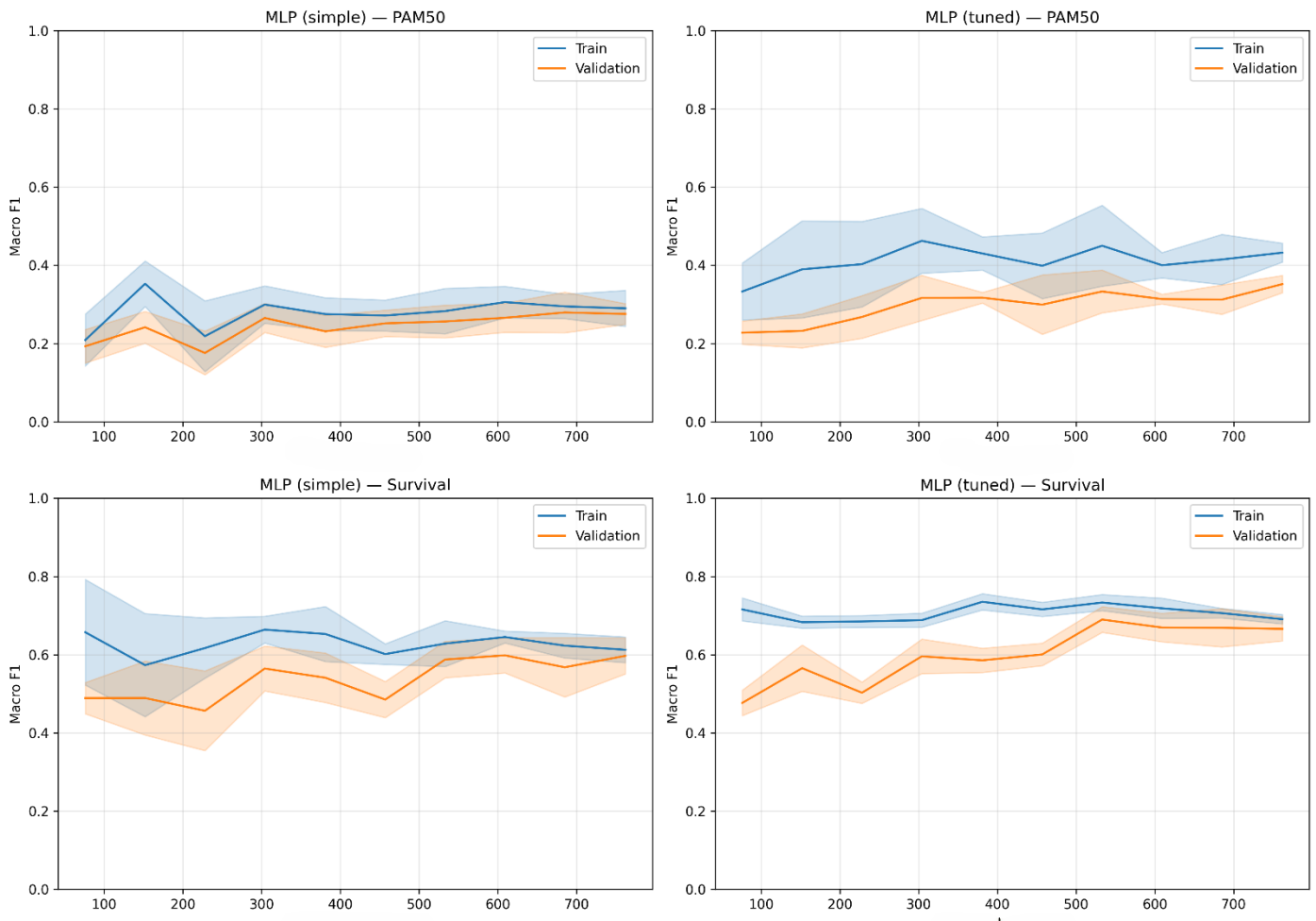


Figure 6: Training and validation curves converge, indicating no overfitting.

For PAM50, both training and validation performance remain relatively modest as the training set size increases. Although the tuned MLP achieves higher scores than the simple architecture, the gap between training and validation performance persists, suggesting limited gains from increased model complexity. Overall, the learning curves indicate that the MLP is unable to substantially improve upon the information already accessible through simpler models.

In contrast, the survival task exhibits a more favourable learning behaviour. Training and validation curves increase together while remaining relatively close, indicating good generalisation and limited overfitting. This observation is consistent with the stronger predictive signal present in the Level 3 embeddings.

6.2 SUMMARY OF RESULTS

For PAM50 classification, the best MLP configuration (tuned MLP on Level 2(GR-DAE)) achieved a Macro F1-score of 0.334 ± 0.021 . Although this represents a clear improvement over Levels 0, 1, and 3, it remains substantially below the performance obtained by the tuned Logistic Regression model (0.433 ± 0.010).

For survival prediction, the tuned MLP on Level 3(GS-DAE) achieved a Macro F1-score of 0.671 ± 0.012 , approaching the performance of the best simple models but not exceeding them.

As observed throughout the previous experiments, stage prediction remained challenging, with all MLP configurations producing Macro F1-scores below 0.20.

6.3 INTERPRETATION

The MLP results provide additional evidence that the learned embeddings already expose much of the clinically relevant information through relatively simple decision boundaries. For PAM50 classification, neither increased model complexity nor non-linear transformations were able to outperform the best linear models. This suggests that the discriminative structure of the Level 2(GR-DAE) embeddings is largely linearly accessible.

For survival prediction, the MLP achieved competitive performance, confirming that the Level 3(GS-DAE) embeddings contain meaningful prognostic information. However, the results were still

obtained by simpler approaches such as Linear SVM, Logistic Regression, and k-NN. To ensure that these observations were not an artifact of suboptimal architectural choices or training parameters, a systematic hyperparameter optimization phase was executed.

6.4 SYSTEMATIC HYPERPARAMETER OPTIMIZATION FOR THE MLP

To fully exploit the capacity of neural networks and address the limitations of initial manual tuning, a comprehensive hyperparameter search was performed for the MLP using GridSearchCV with 5-fold stratified cross-validation.

The grid search explored various architectures and regularisation strategies, including:

- Number of hidden layers and units per layer (*hidden_layer_sizes*)
- Learning rate (*learning_rate_init*)
- Batch size (*batch_size*)
- Regularisation parameter (*alpha*)

For PAM50, the optimal architecture was found to be a single hidden layer with 50 units, a batch size of 32, and a learning rate of 0.01. This configuration yielded a Macro F1-score of 0.3859 on Level 2 embeddings.

For survival prediction, the best configuration consisted of two hidden layers (128 and 64 units), a batch size of 16, and a learning rate of 0.001, achieving a Macro F1-score of 0.6943 on Level 3 embeddings.

Overall, even after rigorous optimization, the MLP did not outperform the best simple models for any prediction task (remaining at 0.386 vs 0.433 for PAM50, and 0.694 vs 0.705 for survival). These findings strongly reinforce the conclusion that embedding quality plays a larger role than classifier complexity in determining downstream predictive performance, as the clinically relevant information encoded in the representations is predominantly linearly accessible.

7 STATISTICAL VALIDATION OF REPRESENTATION LEVELS

To rigorously validate the observed performance differences between embedding levels, paired t-tests were performed on the macro F1-scores obtained over the 10 runs. The analysis was focused on the best-performing configurations among the simpler models: Logistic Regression ($C=10$) for PAM50 classification and Linear SVM ($C=10$) for survival risk prediction.

- **PAM50 Classification:** The improvements from Level 0 to Level 2 ($t = 75.41$, $p < 0.001$) and from Level 1 to Level 2 ($t = 42.64$, $p < 0.001$) were statistically significant, mathematically confirming the benefit of incorporating PPI graph regularisation for this task. Conversely, the drop in performance from Level 2 to Level 3 was also highly significant ($t = -74.05$, $p < 0.001$), while the difference between Level 3 and the Level 0 baseline was not ($t = -1.14$, $p = 0.2839$). This validates the hypothesis that the GNN encoder does not provide a meaningful advantage for PAM50 classification, likely due to the over-smoothing of fine-grained molecular signatures during message passing.
- **Survival Prediction:** For this target, only the comparisons involving Level 3 reached statistical significance. The performance improvements from Level 2 to Level 3 ($t = 21.48$, $p < 0.001$) and from Level 0 to Level 3 ($t = 23.51$, $p < 0.001$) confirm that the GNN architecture is statistically necessary to capture complex prognostic information. In contrast, the differences between the lower representation levels (Level 0, Level 1, and Level 2) did not achieve statistical significance ($p > 0.05$), meaning they do not differ significantly from one another on this task.

8 FINAL SUMMARY TABLE

The table below summarises the best performance obtained for each clinical target across all evaluated embedding levels and classifiers. Results are reported as mean $\pm$ standard deviation over 10 repetitions.

Target	Best level	Best model	F1 (mean $\pm$ std)	Balanced accuracy (mean $\pm$ std)	AUC (mean $\pm$ std)
PAM50	Level 2	Logistic Regression (C=10)	0.433 $\pm$ 0.010	0.437 $\pm$ 0.008	0.792 $\pm$ 0.003
Survival	Level 3	Linear SVM (C=10)	0.705 $\pm$ 0.004	0.706 $\pm$ 0.003	0.763 $\pm$ 0.003
Stage	Level 3	XGBoost	0.227 $\pm$ 0.016	0.228 $\pm$ 0.012	0.575 $\pm$ 0.033

8.1 CLASS-SPECIFIC ERROR ANALYSIS (CONFUSION MATRIX)

To better understand the specific challenges of each prediction task, confusion matrices were generated for the best-performing models identified above. The values represent the aggregated predictions across the cross-validation folds.

PAM50 (Level 2, Logistic Regression, C=10)

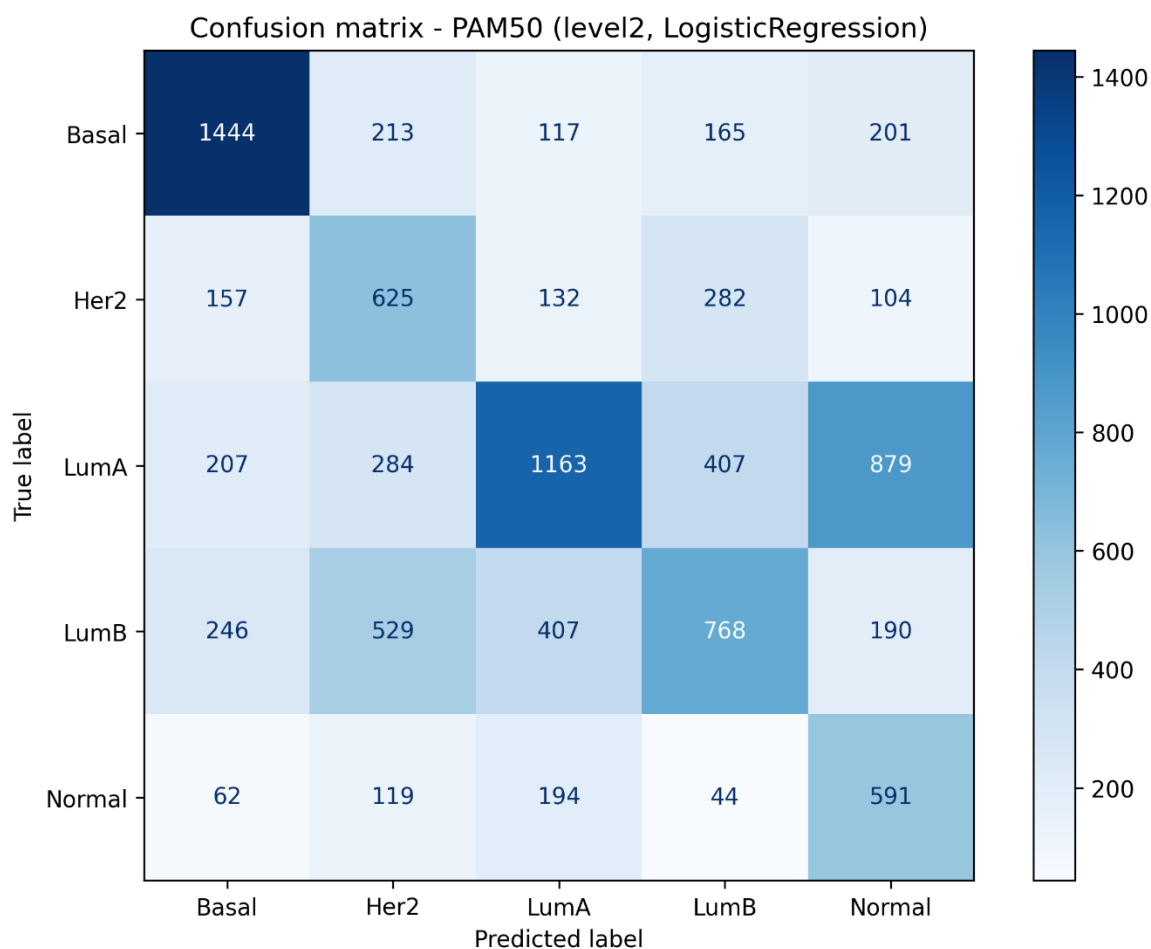


Figure 7 : Confusion Matrix of PAM50.

The confusion matrix for PAM50 reveals that the model performs well for the Basal subtype, with 1,444 correct predictions and relatively few misclassifications. Luminal A is also reasonably well classified (1,163 correct), but it is frequently confused with Normal-like (879 misclassifications), suggesting a biological proximity between these two subtypes. Luminal B shows the poorest performance, with only 768 correct predictions; it is often misclassified as Her2 (529 cases) and Luminal A (407 cases). This confusion likely reflects the molecular overlap between Luminal B and Her2-enriched subtypes. Finally, Normal-like is correctly classified in 591 cases but is also confused with Luminal A (194 cases), which may be due to its small sample size and biological similarity to Luminal A. Overall, the main errors occur between closely related subtypes (Luminal A vs. Normal-like, and Luminal B vs. Her2).

Survival (Level 3, Linear SVM, C=10)

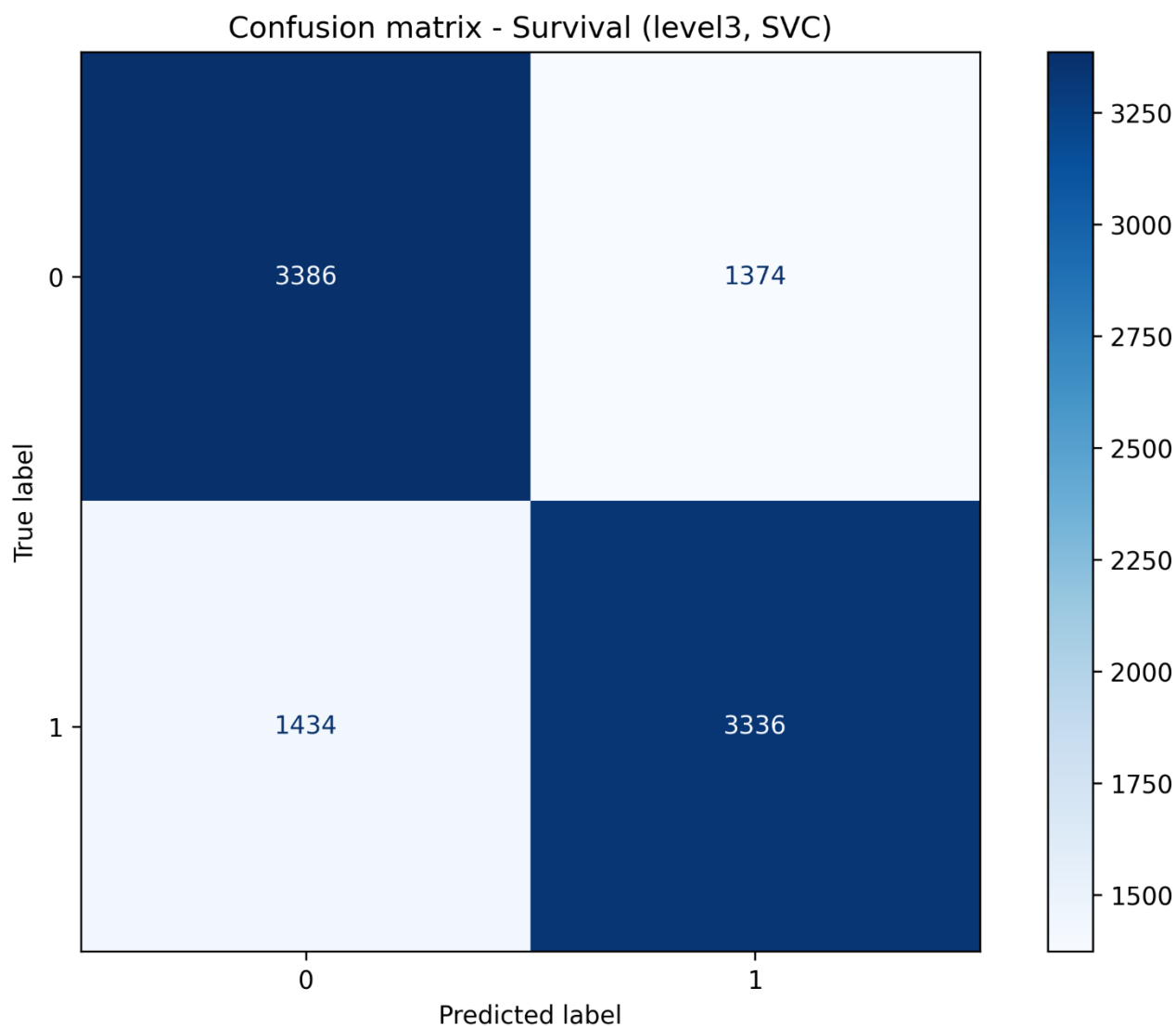


Figure 8 : Confusion Matrix of Survival.

The confusion matrix for survival prediction shows a relatively balanced error distribution between the two classes. Class 0 (low risk) is correctly predicted in 3,386 cases, with 1,374 misclassifications. Class 1 (high risk) is correctly predicted in 3,336 cases, with 1,434 misclassifications. The errors are approximately symmetrical, indicating that the model captures a genuine prognostic signal but does not perfectly separate the two risk groups. This is consistent with the moderate Macro F1-score of approximately 0.70 and suggests that the embeddings contain survival-relevant information that remains partially entangled.

Stage (Level 3, XGBoost)

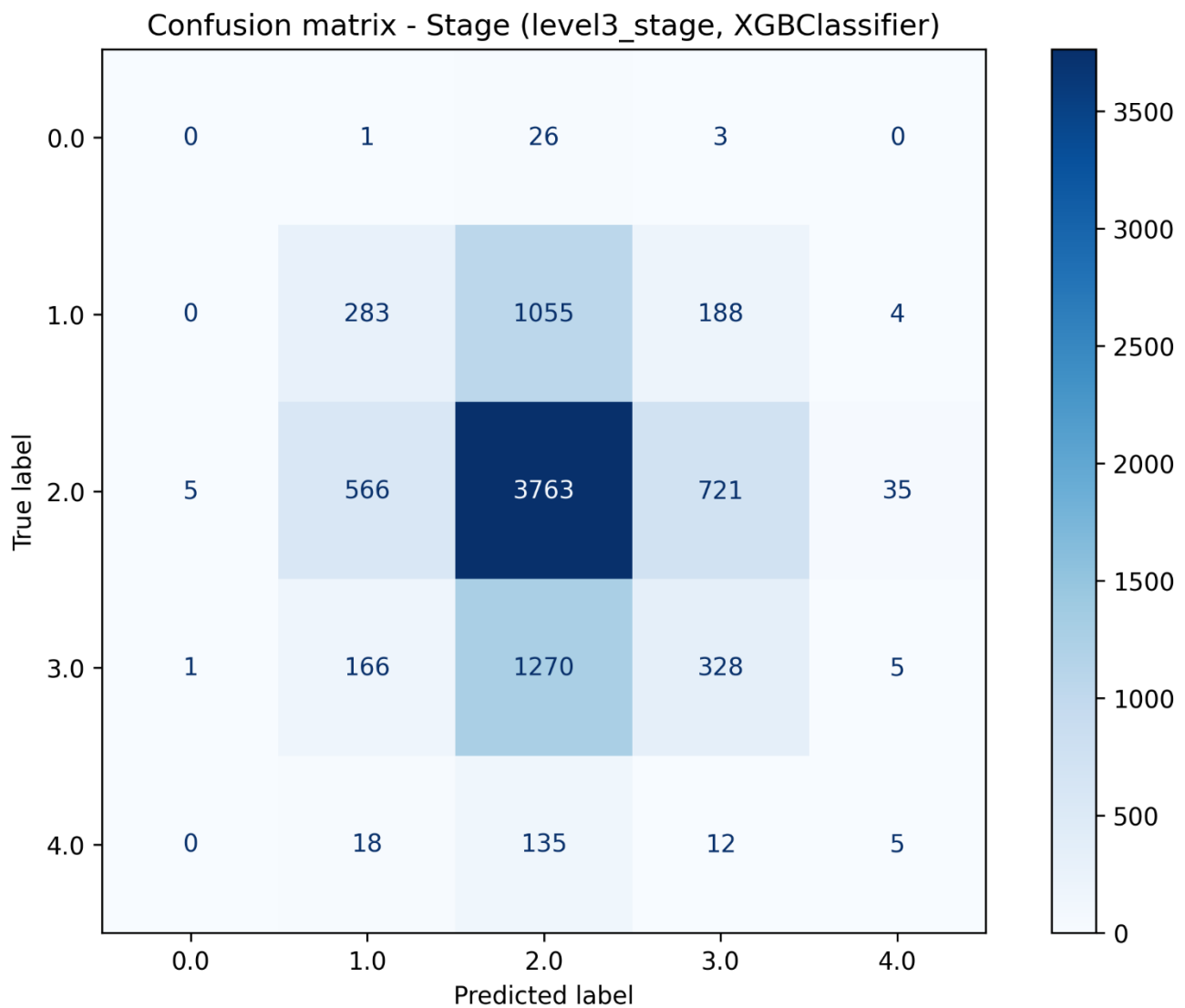


Figure 9 : Confusion Matrix of Stage.

The confusion matrix for stage prediction visually confirms the extreme difficulty of this task. Only Stage II is reasonably well predicted (3,763 correct predictions). Stage 0 is never correctly predicted, and Stage IV is correctly predicted only 5 times, directly reflecting the extreme class imbalance in the dataset. Most predictions are concentrated in Stage II, which is the majority class, indicating that the model defaults to predicting the most frequent class rather than capturing the underlying ordinal structure of tumour stages. This behaviour suggests that the embeddings contain limited stage-related signal, and that the severe imbalance further hinders the learning process.

8.2 MAIN FINDINGS

The results demonstrate that different architectural choices favour different types of clinical information.

For PAM50 classification, the best performance is obtained with the Level 2(GR-DAE) embeddings generated by the graph-regularised denoising autoencoder (GR-DAE). Logistic Regression with C = 10 achieves a Macro F1-score of 0.433 ± 0.010 , outperforming all other classifiers, including ensemble methods and neural networks. These findings suggest that incorporating PPI information through graph regularisation improves the biological coherence of the latent space and facilitates the separation of molecular subtypes.

For survival prediction, the best performance is achieved using Level 3(GS-DAE) embeddings produced by the GNN encoder. A Linear SVM reaches a Macro F1-score of 0.705 ± 0.004 , substantially outperforming the lower representation levels. This indicates that graph propagation captures biologically meaningful information that is particularly relevant for prognostic prediction.

Stage prediction remains challenging regardless of the representation level or classifier. The highest Macro F1-score observed is 0.227 ± 0.016 , suggesting that stage-related information is either weakly represented in the embeddings or difficult to recover given the severe class imbalance present in the dataset, particularly for Stage 0 and Stage IV.

8.3 COMPARISON OF REPRESENTATION LEVELS

The comparison across embedding levels reveals distinct strengths and weaknesses.

Level	Main Strength	Interpretation
Level 2	PAM50	PPI regularization improves subtype-related information

Level 3	Survival prediction	Graph propagation enhances prognostic information
Level 0-1	Limited clinical signal	Weak predictive performance across all tasks

The progression from Level 1(DAE) to Level 2(GR-DAE) provides partial validation of the initial hypothesis regarding graph regularisation. For PAM50 classification, performance improves substantially, with the best Macro F1-score increasing from approximately 0.20 to 0.43. In contrast, no comparable improvement is observed for stage prediction. These results suggest that graph regularisation primarily benefits the encoding of molecular subtype information rather than disease progression-related signals.

The transition from Level 2(GR-DAE) to Level 3(GS-DAE) reveals a different trade-off. While survival prediction improves considerably, PAM50 performance decreases. One possible explanation is that graph propagation attenuates fine-grained molecular signatures by making neighbouring genes more similar in the latent representation. Such an effect would be detrimental for subtype discrimination while remaining beneficial for broader prognostic patterns.

8.4 ANALYSIS OF STAGE PREDICTION

To investigate whether the poor performance observed for stage prediction reflects a genuine absence of signal or is primarily driven by class imbalance, two complementary analyses were conducted.

First, the AUC for stage prediction on Level 3 embeddings ranges between 0.59 and 0.62, consistently above the random baseline of 0.50. This suggests that the embeddings do contain a weak but detectable ordinal signal that is not captured by the Macro F1-score, which is heavily penalised by the very small number of Stage 0 and Stage IV samples.

Second, a binary version of the stage task was constructed by grouping Stage 0, I, and II as “early” (0) and Stage III and IV as “advanced” (1). This simplification partially mitigates the class imbalance issue. Using a balanced logistic regression on Level 2 embeddings, the binary classifier achieved a Macro F1 of 0.50. The per-class analysis revealed that the advanced group (III-IV) had a recall of 0.61 but a precision of only 0.27, indicating that the model tends to over-predict advanced stages. The early group (0-I-II) showed a higher precision (0.82) but a lower recall (0.52), meaning that 321 out of 665 early-stage patients were misclassified as advanced.

These results confirm that while a weak ordinal signal exists, it is not sufficient for reliable stage prediction. The severe class imbalance, combined with limited stage-related information in the embeddings, makes stage prediction a challenging task even in its simplified binary form.

CONCLUSION

This study successfully demonstrates that deep patient representations learned from transcriptomic data can capture critical clinical dimensions of breast cancer, provided that the architectural inductive biases are aligned with the nature of the clinical target. Rather than identifying a single universally optimal embedding, the results reveal a fundamental biological trade-off between graph regularisation and graph propagation, each serving distinct clinical utilities.

Level 2 embeddings (incorporating PPI graph regularisation) proved highly effective for predicting PAM50 molecular subtypes. Clinically, this indicates that constraining the latent space to respect known protein interaction networks preserves the localized, distinct molecular signatures that define breast cancer intrinsic subtypes (Basal, Her2, Luminal A/B). Because PAM50 subtyping is a cornerstone for selecting targeted therapies (such as endocrine therapy or anti-HER2 agents) in clinical practice, Level 2 representations offer a robust basis for patient stratification and therapy selection.

Conversely, Level 3 embeddings (utilising a GNN encoder with message passing) provided the strongest prognostic signal for overall survival risk. From a medical perspective, survival is a complex, systemic phenotype driven by broad, multi-pathway biological disruptions rather than a single set of marker genes. While information propagation across the PPI network attenuates fine-grained subtype signatures due to over-smoothing, it successfully synthesises the global network disruptions that influence long-term prognosis. Consequently, Level 3 embeddings provide an effective foundation for long-term risk assessment and survival forecasting.

The persistent difficulty in predicting pathological stage across all levels, despite a weak underlying ordinal signal revealed by the AUC metrics, highlights that anatomical tumor progression may not be strictly coupled with basal gene network topology, or that severe data imbalance masks this relationship. Finally, the finding that optimized, highly non-linear classifiers (Ensembles and MLPs) failed to consistently outperform simple linear models carries crucial clinical weight: it implies that the clinically relevant information within the embeddings is already linearly accessible. For future translation into clinical decision-support systems, this linear accessibility is ideal, as it allows the deployment of simpler, more interpretable, and computationally transparent models that can easily win the trust of medical practitioners.

In conclusion, this work demonstrates how network-derived patient embeddings can capture clinically meaningful information and highlights the importance of aligning architectural inductive

biases with the nature of the clinical target. Future work will focus on the biological interpretation of the latent dimensions and on the validation of the embeddings through patient retrieval.